MAXIMILIAN WOJCIECHOWSKI

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EDUCATION

DePaul University — **Chicago, IL**
Master of Science in Computer Science; GPA: 3.8/4.0 — *Expected March 2028*
- Research interests: Trading, machine learning for financial markets, and AI engineering
- Relevant Coursework: Data Structures I–II, Computer Systems I–II, Discrete Mathematics, Python Programming

University of New Orleans — **New Orleans, LA**
Bachelor of Science in Finance, Data Analytics Concentration — *May 2025*
- Athletic and academic scholarship; Privateer Blue Honors (Fall 2022, Spring 2023, Spring 2024)
- Relevant Coursework: Investments, Financial Modeling, Quantitative Methods, Data Analytics, Accounting

Wichita State University — **Wichita, KS**
Bachelor of Arts in International Business — *Aug 2020 – May 2022*
- Athletic and academic scholarship; NCAA All-Academic Team (2021–22)

EXPERIENCE

DekaBank — **Frankfurt, Germany**
Intern, Structuring & Derivatives Trading — *Jun 2026 – Sep 2026*
- Developed a regex-based Python algorithm that classifies thousands of dpa news articles by issuer, category, and country and maps them to client portfolios via LEI/ISIN/WKN; currently being migrated into production
- Built a backtesting engine to validate the classifier's output against expected issuer and portfolio mappings
- Built a Java rule engine automating counterparty classification under EU CRR, encoding 40+ bank-specific rules, with a test suite validating edge cases and rule priorities
- Code-reviewed and performance-tested an LLM-based term-sheet comparison tool, systematically validating the quality and accuracy of its outputs (Python)
- Built an interactive world map prototype for the internal trading dashboard with Claude (TypeScript), combining internal financial data with external economic indicators; presented to senior engineers
- Wrote internal wiki documentation on AI-assisted and agent-based coding workflows, using the world map as the case study

SWOT Controlling GmbH — **Berlin, Germany**
Intern, Financial Reporting & Systems — *Jun 2025 – Aug 2025*
- Built P&L and consolidation reports, standardizing financial data across five companies
- Translated controlling software from German to English; supported system integration and client rollout alongside the CEO

Schwanke & von Schwerin — **Berlin, Germany**
Intern, Tax Accounting & Audit — *May 2024 – Aug 2024*
- Conducted audit preparation and document reviews, reducing filing errors by 15%; flagged 10% budget cost variances
- Automated Excel checks that caught input errors from company software, improving reporting efficiency

PROJECTS

Local RAG & Evaluation Pipeline — **Python, MLX, Llama 3.1 8B, Chroma**
- Built a fully local RAG pipeline over FY2025–26 10-K filings of 5 companies (Microsoft, NVIDIA, Tesla, Lockheed Martin, BlackRock), using hybrid dense + BM25 retrieval with no external API calls
- On 65 hand-verified questions: 0.87 answer score vs. 0.24 without retrieval (95% bootstrap CIs), 0.92 hit@5; oracle tests showed generation, not retrieval, was the bottleneck

Macroeconomic Forecasting System — **TypeScript, World Bank/EIA APIs, Claude tool use**
- Built a live economic and geopolitical dashboard on 15+ data streams, with in-browser forecasts and a 3D world map
- Added an AI analyst that answers by calling 25 live-data tools (World Bank, EIA, ACLED, market quotes)
- Chose models by rolling-origin backtest across 217 economies (1990–2025) under a pre-specified rule: AR(2) cut GDP error 8% (1-yr) and 20% (3-yr) vs. no-change but lost on inflation and unemployment (now no-change); IMF beat AR(2) in 142 of 185
- 95% intervals calibrated pre-2010 covered 93–96% of 2010–25 outcomes; weekly US natural-gas storage forecasts beat the five-year seasonal norm by 5% over 561 weeks (Diebold–Mariano p = 0.004; 80% intervals covered 81%)

NEOTracker — **Swift, SwiftUI, SceneKit, NASA APIs**
- iOS app that turns NASA near-Earth-object data into real-time close-approach tracking with 3D orbits, plus a GPS-based night-sky planner computing planet positions and rise/set times (Meeus/VSOP87 algorithms)

SKILLS & INTERESTS

Languages: German (native), English (C2), Polish (C2)
Programming: Python, Java, TypeScript, Swift, C, Git, Claude Code; RAG (Chroma, BM25, MLX)
Business: Excel (financial modeling), PowerPoint, Bloomberg Terminal
Activities: NCAA D1 Golf (5 years); German National Golf Team (2018–2020); DePaul Computer Science Society
Interests: Golf, geopolitics, cosmology, trading